FLORIAN BANON

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Website – LinkedIn – GitHub – Projects

Aerospace Engineer specialised in **Space, Launchers & Satellites**, completing a dual Master's between IPSA (France) and Concordia University (Canada). My core focus is **orbital mechanics**, **interplanetary mission analysis** and **numerical simulation**, combined with **data engineering** and **machine learning**. I'm currently finishing a research internship at NUS, building physics-informed CVNNs to mitigate ionospheric scintillation in SAR imagery — with a paper targeting ISAP 2026 — and I'm now seeking a full-time opportunity from August 25, 2026.

SKILLS

Orbital Mechanics & Astrodynamics | Orbital propagation • Perturbation analysis • Interplanetary mission design (patched-conic, gravity assist) • ΔV & mass budgets • Monte Carlo uncertainty propagation • Orbit determination & estimation theory (Kalman, least-squares — coursework)
Machine Learning & Deep Learning | PyTorch • Complex-valued neural networks (CVNN) • CNNs • Physics-informed deep learning • SAR imagery • HPC pipelines (PBS, GPU)
Programming| Python (advanced) • MATLAB • C/C++ • SQL • R • Bash • Git • VBA • LaTeX
Space & Engineering Software | STK (AGI Systems Tool Kit) • Orekit • Skyfield • MongoDB • Star-CCM+ • ANSYS • CATIA V5 • SolidWorks
Data Engineering | Data ingestion & preprocessing pipelines • Large-scale / high-dimensional datasets • HPC (PBS, conda)

EDUCATION

Master of Aerospace Engineering – Co-op **2024-2026**
Concordia University, Montreal, QC

Master of Aerospace Engineering **2022- 2026**
Institut Polytechnique des Sciences Avancées (IPSA), Paris, France
- Specialization in Space, Launchers and Satellites

WORK EXPERIENCE

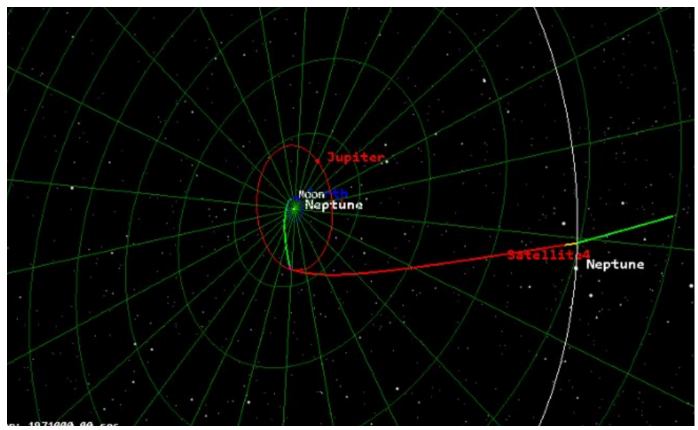
Research Engineering Intern – Deep Learning for Ionospheric Scintillation Mitigation in SAR Imagery
Jan 2026 - present
National University of Singapore, Singapore
- Designed a **complex-valued neural network (CVNN)** predicting scintillation indices (S4, SS) from complex SAR data; $R^2 = 0.86$, beating random-forest and real-valued baselines.
- Built a **physics-informed pipeline** (CVNN → ionospheric transfer function → U-Net denoiser) for restoring L-band SAR imagery; paper targeting **ISAP 2026**.
- Engineered the **PyTorch** training stack on an **HPC cluster (PBS, GPU)**: complex HDF5 I/O, precompute caching (~5× faster training), per-output metrics.
- Coordinated the **data interface with the physics team** and authored the project's technical documentation.

Software & Data Engineering Intern – Dynamic Satellite Database for Space Surveillance.
Jun 2024 – Aug 2024
ONERA (French Aerospace Lab), Palaiseau, France

- Built OBELIX, a **Python ETL pipeline** for space-surveillance data, aggregating 4 heterogeneous sources (Gunter, UCS, TLE, DORIS/CNES) via web scraping.
- Derived orbital parameters with **Skyfield** and classified orbits (LEO/MEO/GEO/SSO…), incl. sun-synchronous detection via J2 precession.
- Centralised the catalogue in MongoDB (dedup + integrity checks) and exposed it through a **FastAPI REST API**.
- Documented the schema and procedures so the system could be maintained and extended by the team.

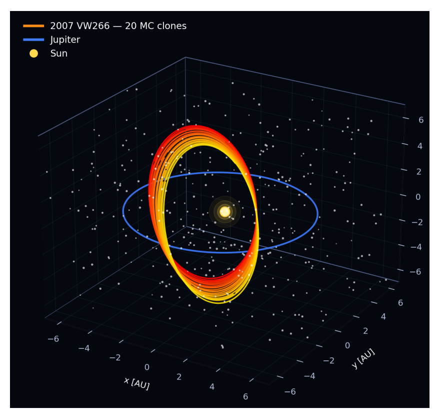
RELEVANT SPACE PROJECTS

Neptune Flyby Mission



Group project, IPSA

- Designed an Earth–Jupiter–Neptune **patched-conic trajectory** with a single Jupiter gravity assist, using the **NASA Trajectory Browser** solution as the design baseline.
- Analytically computed each conic segment — circular parking orbit, hyperbolic Earth escape (≈ 7.92 km/s), Earth→Jupiter transfer ellipse, Jupiter gravity-assist geometry (≈ 11.6 km/s free heliocentric gain), and Jupiter→Neptune hyperbolic transfer.
- Built the **ΔV & mass budgets** (Tsiolkovsky) and reproduced the full trajectory in an **STK12** simulation, consistent with the NASA reference (~7 yr).



Dynamical Simulation of the Asteroid Population

Group project, IPSA

- Modelled the motion of asteroid 2007 VW266 in 13/14 retrograde resonance with Jupiter; deriving the equations of motion from Keplerian to Cartesian coordinates with Sun and Jupiter perturbations.
- **Python RK4 propagation** + **Newton-Raphson Kepler solver**.
- **Monte Carlo uncertainty propagation** over 10,000 yr to study the evolution of the orbital elements; cross-validated against the Connors and Wiegert reference paper.

ADDITIONAL

Volunteer: Ropes Course Instructor — safety-critical procedures, clear communication, attention to detail.
Interests: Bouldering (3 yrs), Tennis (5 yrs), Chess (1300 Glicko).